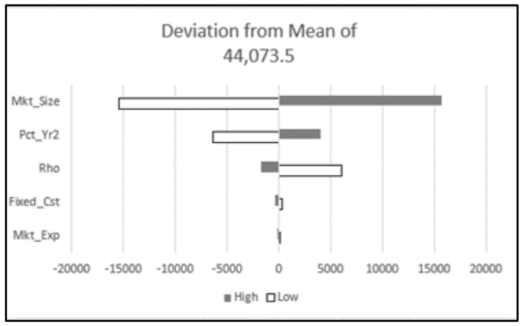
Sensitivity Analysis for ChanceCalc



By Dave Empey

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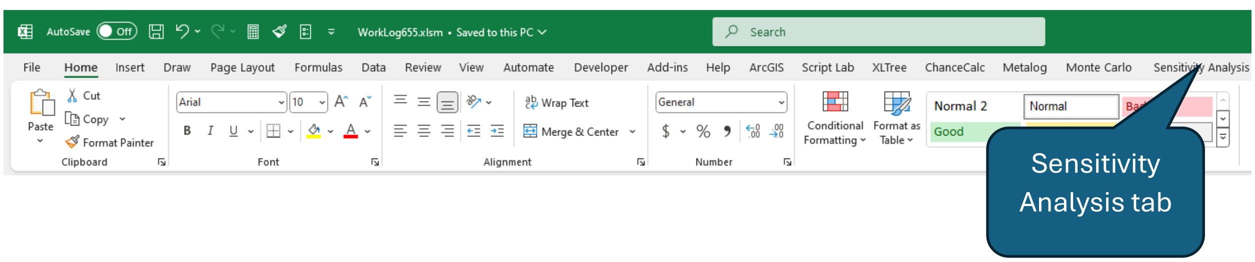
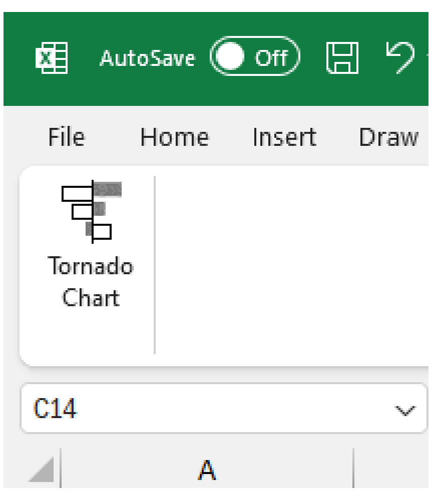
NOTE: If you have a ChanceCalc or SIPmath model with a PMTable set up, this will overwrite your existing PMTable!

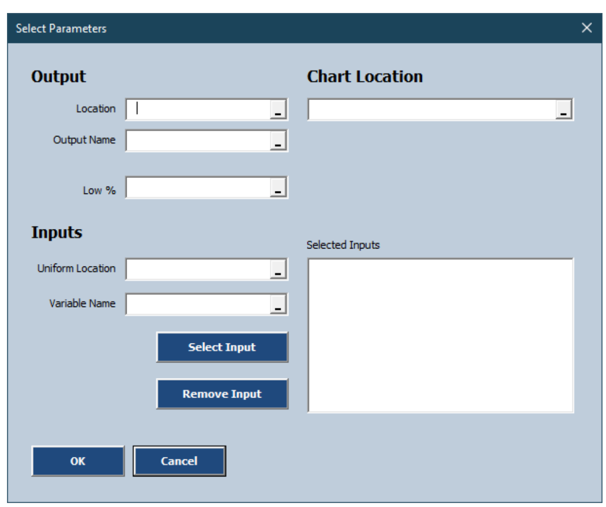
The Sensitivity Analysis tool is found under the tab named Sensitivity Analysis on the Excel Ribbon.



Click the tab to show the Tornado Chart sensitivity analysis button.

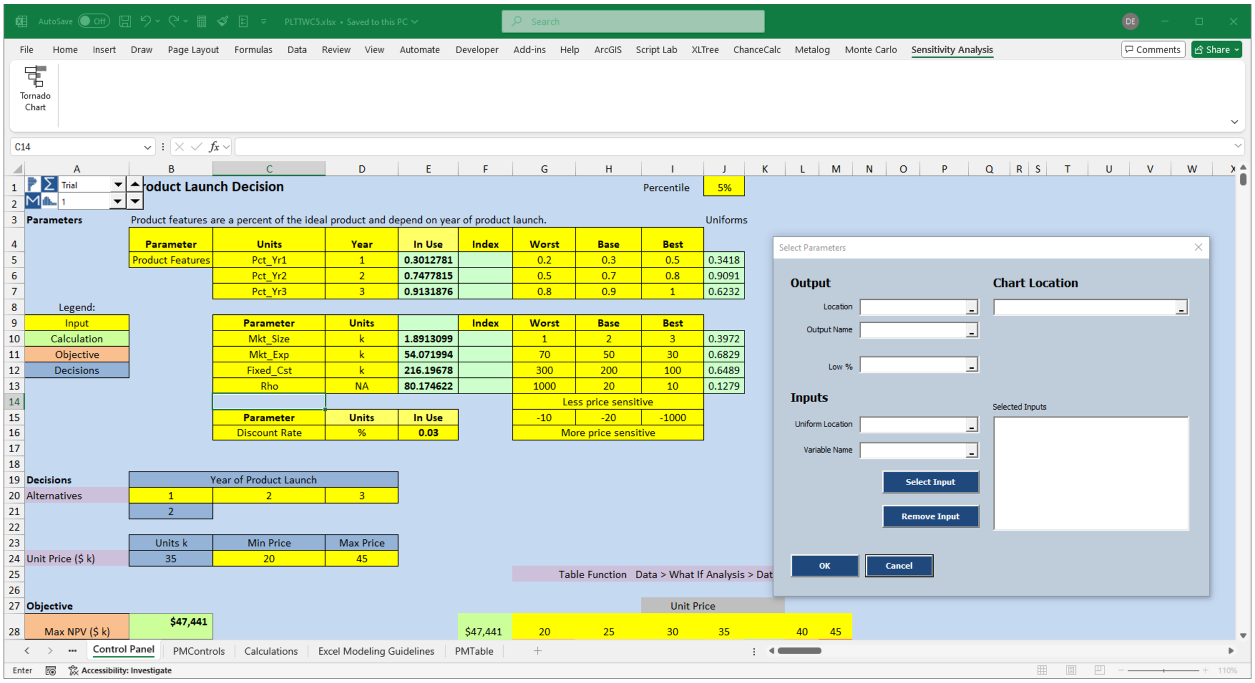


Press the Tornado Chart button to open the Tornado Chart parameter entry form.



Note: if you select a cell with a formula before pressing the button, that cell address will be filled in as the Output cell Location.

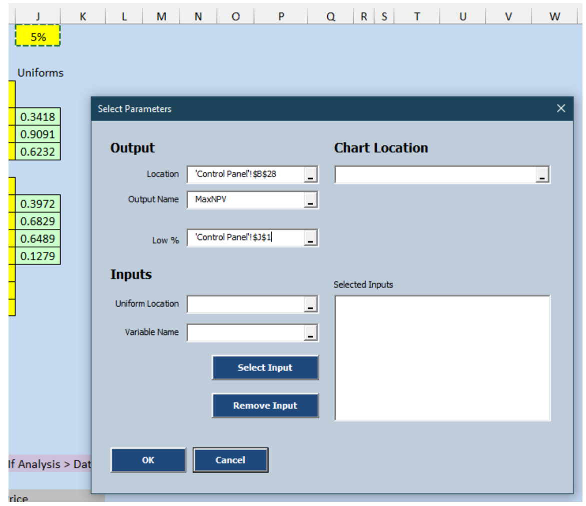
Click in the Output Location field, and then select the cell containing the formula whose sensitivity you want to analyze. The Output Location field will be filled in with the cell address. In the example below, the Output Location cell is ‘Control Panel’!B28.



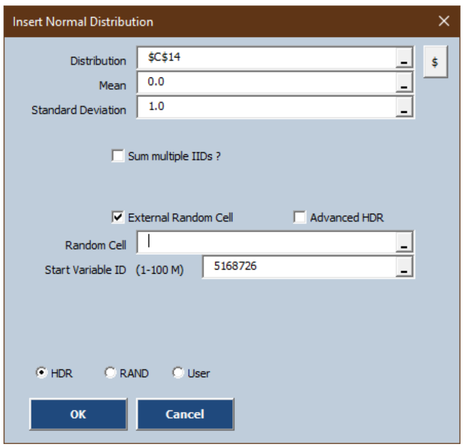
Click in the Output Name field. You can either click on a cell containing the Output Name, or type in the Output Name you want. The name should consist of letters, numbers, and underscore characters and have no spaces.

In Tornado Chart sensitivity analysis, you vary a set of Inputs by setting each one successively to a low value, perhaps the 5th percentile of the input range, and then to a high value, which for this tool will be 100%-the low value %.

Click the Low% field and then click the cell with the low % value you want to use for your sensitivity analysis.



Input variables:



With this tool, Input Variables must be determined by External Uniform random variables, such as are created by the SIPmath Generate Input menu option; for example the Normal Distribution.

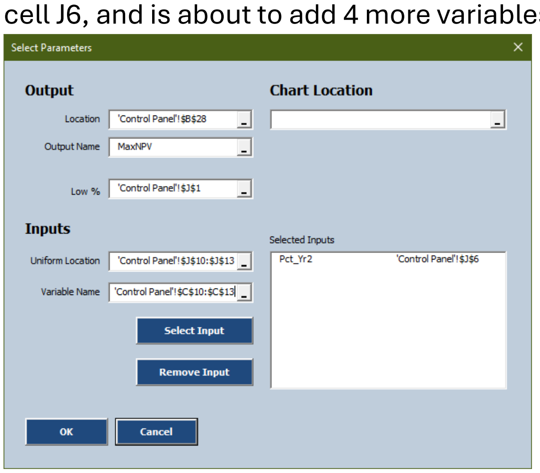
That is, you must choose the External Random Cell checkbox, as shown in the example here.

Selecting Input Variables:

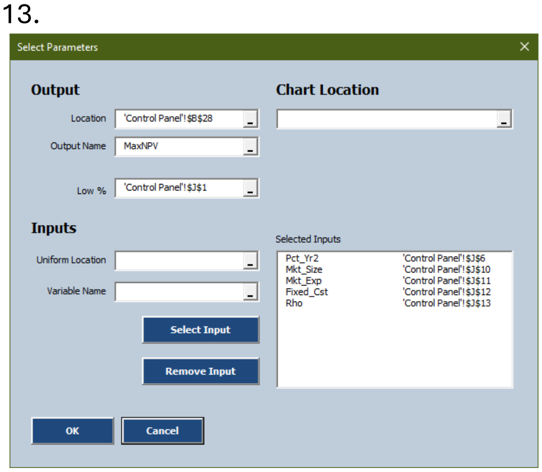
Since typically you will want to determine the sensitivity to several different input variables, the Probability Management Tornado Chart tool allows you to create a list of several variables to analyze.

To add Input Variables to the list, select the Uniform Location (the External Uniform cell mentioned above), the Variable Name, and press the Select Input button on the form.

If you have a range of input variables, you can add them all at once by selecting them (and their names) and pressing the Select Input button. Here the user has already added an input variable at cell J6, and is about to add 4 more variables at cells J10:J13.



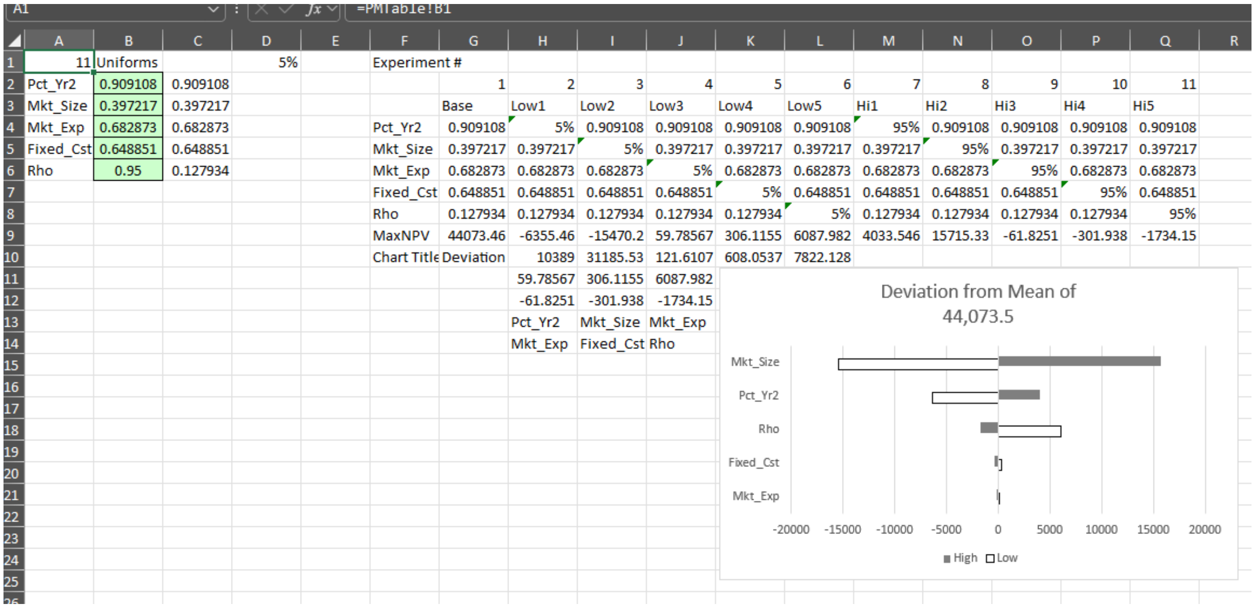
The cells have been added.



If you make a mistake adding input cells, you can remove the mistaken entry by clicking the mistaken entry in the list of selected inputs and pressing the Remove Input button.

You can specify a location for the tornado chart in the Chart Location cell; if you leave this blank, the chart will be created on a new page called PM_Sensitivity created by the tool.

When you have selected all your sensitivity parameters, press the OK button and the tool will create a tornado chart based on the model and the cell and % values you entered.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	11	Uniforms		5%		Experiment #												
2	Pct_Yr2	0.909108	0.909108				1	2	3	4	5	6	7	8	9	10	11	
3	Mkt_Size	0.397217	0.397217				Base	Low1	Low2	Low3	Low4	Low5	Hi1	Hi2	Hi3	Hi4	Hi5	
4	Mkt_Exp	0.682873	0.682873			Pct_Yr2	0.909108	5%	0.909108	0.909108	0.909108	0.909108	95%	0.909108	0.909108	0.909108	0.909108	
5	Fixed_Cst	0.648851	0.648851			Mkt_Size	0.397217	0.397217	5%	0.397217	0.397217	0.397217	0.397217	95%	0.397217	0.397217	0.397217	
6	Rho	0.95	0.127934			Mkt_Exp	0.682873	0.682873	0.682873	5%	0.682873	0.682873	0.682873	0.682873	95%	0.682873	0.682873	
7						Fixed_Cst	0.648851	0.648851	0.648851	0.648851	5%	0.648851	0.648851	0.648851	0.648851	95%	0.648851	
8						Rho	0.127934	0.127934	0.127934	0.127934	0.127934	5%	0.127934	0.127934	0.127934	0.127934	95%	
9						MaxNPV	44073.46	-6355.46	-15470.2	59.78567	306.1155	6087.982	4033.546	15715.33	-61.8251	-301.938	-1734.15	
10						Chart Title	Deviation	10389	31185.53	121.6107	608.0537	7822.128						
11								59.78567	306.1155	6087.982								
12								-61.8251	-301.938	-1734.15								
13								Pct_Yr2	Mkt_Size	Mkt_Exp								
14								Mkt_Exp	Fixed_Cst	Rho								

This chart is live, in that it will update when parameters are changed within the model.